Koa Shen

TEST ENGINEERING | SUMMER 2027

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PROFESSIONAL SUMMARY

Mechanical engineering student with hands-on experience validating electromechanical hardware, inspecting precision-machined parts, qualifying mechanical load cases, and diagnosing field-use failures. Combines structured engineering analysis with practical bench testing, measurement, and clear documentation.

EDUCATION

University of California, Santa Barbara | B.S. Mechanical Engineering
Santa Barbara, CA | Expected June 2028 | Major GPA: 3.97 | Cumulative GPA: 3.94
Dean's Honors every quarter | Relevant coursework: Mechanics of Materials, Dynamics, Circuits, MATLAB

TEST AND ENGINEERING EXPERIENCE

Mechanical Engineering Intern | SLAC National Accelerator Laboratory, LCLS
Summer 2026–Present | Menlo Park, CA; extended part-time remote

- Built a Python/Drake digital twin for a 30-DOF, 305-body beamline assembly; live collision checks complete in milliseconds and exposed at least five real pinch points for engineering review.
- Performed statics and dynamics hand calculations to evaluate stage and mount loads against crash and drop cases involving detectors valued at more than $100,000.
- Produced Siemens Solid Edge designs and Teamcenter PDM-controlled drawings with GD&T to communicate manufacturing and inspection requirements.

Fabrication Team Lead | Gaucho Racing, Formula SAE Electric, UC Santa Barbara
Fall 2024–Present | Santa Barbara, CA

- Verified precision-machined suspension features using go/no-go tests; held critical bores within a 0.0004-in tolerance window and bearing interfaces within 0.0005 in.
- Led 15 teammates through driver-safety-system design, fabrication, and vehicle installation; complete GR26 package passed Formula SAE technical inspection.
- Used iterative design and FEA-guided development to reduce steering-wheel mass from 5 lb to 1.7 lb while integrating controls and quick-release hardware.

Maintenance Technician | Exploratorium
Summer 2025 | San Francisco, CA

- Diagnosed and repaired mechanisms across 100+ public exhibits; reverse-engineered and machined replacement parts for hardware subjected to continuous, unpredictable use.
- Redesigned encoder mounts to address sand ingress and wear; redesigned a load-spreading bezel to address acrylic cracking and avoid $3,000+ in replacement costs.
- Documented CAD and remanufacturing instructions to support repeatable maintenance and future repair.

SELECTED HARDWARE TEST PROJECT

Closed-Loop Cycloidal Actuator | Personal Project

- Designed and fabricated a 15:1 cycloidal reducer packaged within a NEMA 17 motor's 42 mm square footprint.
- Built C++ firmware for a TMC2209 stepper driver and AS5600 encoder; bench-tested single-joint position correction and missed-step detection.

TECHNICAL SKILLS

Test and Verification: Bench testing, go/no-go inspection, precision measurement, load-case analysis, hardware troubleshooting, technical inspection
Mechanical: SolidWorks, Solid Edge, GD&T, FEA, CNC/manual machining, assembly, welding
Software and Controls: Python, C/C++, MATLAB, Drake, PlatformIO, stepper control, encoders, Git